

Ansh Verma

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Summary

ECE student focused on AI hardware acceleration, FPGA systems, and RTL architecture design. Experienced in developing FPGA-based neural processing units (NPUs), systolic-array accelerators, and graph-based ML models for topology-aware power prediction, with interests in computer architecture, hardware-aware machine learning.

Skills

RTL Design — Verilog, FSM Design, Testbench Development, Systolic Arrays | **Computer Architecture & AI Hardware** — FPGA-Based NPUs, CNN Acceleration, RISC-V, Parallel MAC Architectures | **Programming** — Python, Embedded C, Java | **Machine Learning** — Graph Neural Networks (GNNs), Random Forest, Neural Networks, Feature Engineering | **Tools** — Xilinx Vivado, ModelSim, Cadence Virtuoso, LTSpice

Education

Vellore Institute of Technology - Chennai

August 2023 – Present

Bachelor of Technology, Electronics and Communication Engineering
CGPA: 8.09/10.00

Projects

FPGA-Based Mini NPU using Systolic Array Architecture [GitHub] [🔗](#)

May 2026

- Designed a 4×4 FPGA-based Neural Processing Unit (NPU) in Verilog using INT8 MAC units and weight-stationary systolic-array architecture for CNN acceleration.
- Built an RTL-to-bitstream accelerator pipeline with FSM-controlled execution, im2col-based convolution lowering, timing closure (WNS +0.153 ns), and DSP48E1 inference in Vivado.

Gate-Level Power Prediction using Graph Neural Networks (GNNs) [GitHub] [🔗](#)

April 2026

- Developed a topology-aware GraphSAGE framework for gate-level power prediction, achieving an R^2 score of 0.8966 compared to 0.1205 using Random Forest models.
- Designed an “Isomer Pair Benchmark” with identical aggregate statistics but different circuit topologies, demonstrating the importance of structural connectivity in topology-aware power estimation using PyTorch Geometric and scikit-learn.

5G-Based Vehicle-to-Everything (V2X) Emergency Coordination System [GitHub] [🔗](#)

March 2026

- Designed a 5G-enabled V2X coordination system supporting 2–3 simultaneous emergency vehicles with priority-based conflict resolution and token negotiation.
- Achieved 98% corridor formation success rate with reduced clearance time (35s vs 60s baseline) and minimized oscillations (2 vs 12) compared to greedy approach.

AI-Powered 6G Beam Prediction System [GitHub] [🔗](#)

May 2026

- Developed a Deep Residual MLP (1.2M parameters) for 6G massive MIMO beam prediction achieving 87.94% Top-1 and 100% Top-5 accuracy on DeepMIMO mmWave dataset (28 GHz).
- Engineered end-to-end ML pipeline retaining 98.66% optimal channel capacity, eliminating exhaustive beam sweeping and reducing signaling overhead by 95%.

Real-Time Target Tracking Laser Turret using Texas Instruments Board [GitHub] [🔗](#)

December 2025

- Developed embedded tracking system using TM4C123GXL with ultrasonic sensing and PWM-based pan-tilt control (0–180° range).
- Achieved ±1 cm distance accuracy with filtered measurements and implemented threshold-based auto-locking (<30 cm) with real-time OLED feedback via I²C.

Certifications

- Fundamentals of Deep Learning – NVIDIA** [🔗](#) (GPU computing, neural networks, model training basics)
- AWS Academy Graduate – Cloud Foundations – Amazon Web Services** [🔗](#) (cloud computing fundamentals, AWS core services, architecture basics, pricing models)